

02YMECH - Solution 10

Assigned for the week of Nov 24, 2025

Solutions

1. At the equator, the Coriolis acceleration is

$$\mathbf{a}_C = -2\boldsymbol{\Omega} \times \mathbf{v},$$

with $\boldsymbol{\Omega}$ horizontal and pointing north.

The apple falls vertically with

$$v_z(t) = -gt.$$

Thus the Coriolis acceleration is purely horizontal (eastward) with magnitude

$$a_C = 2\Omega gt.$$

Horizontal velocity:

$$v_x(t) = \int_0^t 2\Omega g\tau \, d\tau = \Omega gt^2.$$

Horizontal displacement:

$$x(t) = \int_0^t \Omega g\tau^2 \, d\tau = \frac{1}{3}\Omega gt^3.$$

The fall time from height h is

$$t_f = \sqrt{\frac{2h}{g}}.$$

Hence

$$x = \frac{1}{3}\Omega g \left(\frac{2h}{g}\right)^{3/2} = \frac{2\sqrt{2}}{3}\Omega \frac{h^{3/2}}{\sqrt{g}}.$$

With $h = 4$ m, $\Omega = 7.29 \times 10^{-5} \text{ s}^{-1}$,

$$x \approx 2 \times 10^{-4} \text{ m} = 0.2 \text{ mm}.$$

The deviation is negligible. The phrase “the apple does not fall far from the tree” remains valid :)

2. (a) In the rotating frame, the forces along the pipe are due to the centrifugal force. The equation of motion is

$$m\ddot{r} = m\Omega^2 r.$$

(Coriolis force is perpendicular to the pipe and constrained by the walls.)

- (b) Solution:

$$r(t) = Ae^{\Omega t} + Be^{-\Omega t}.$$

Using $r(0) = r_0$, $\dot{r}(0) = 0$,

$$r(t) = r_0 \cosh(\Omega t).$$

- (c) Time to reach $r = R$:

$$R = r_0 \cosh(\Omega t) \Rightarrow t = \frac{1}{\Omega} \cosh^{-1}\left(\frac{R}{r_0}\right).$$

- (d) Although the particle is initially at rest in the rotating frame, it experiences a centrifugal force $m\Omega^2 r$ directed outward. This force does work on the particle, causing radial acceleration and increasing kinetic energy in the rotating frame.